

Neha Sengar

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RESEARCH INTERESTS

Mechanistic interpretability of large vision-language models, training-free visual grounding, AI-driven methods for robot perception & 3D scene understanding, open-vocabulary scene reasoning, real-world deployment in autonomous systems, generalization.

EDUCATION

Ph.D. in Robotics Program

Feb 2023 – Jan 2027 (expected)

Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Republic of Korea

- Advisor: Prof. Dongsoo Har
- Dissertation: “*Mechanistic Interpretability of Attention Heads in Frozen Vision-Language Models for Training-Free Visual Understanding*”
- Research Group: Vehicular Intelligence Lab

M. Tech. in Computer Science & Engineering

Aug 2019 – July 2021

Centre for Advanced Studies, Lucknow, India

B. Tech. in Computer Science & Engineering

Aug 2016 – July 2019

Azad Institute of Engineering & Technology, Lucknow, India

PUBLICATIONS

*Equal contribution †Corresponding author

Under Review

- N Sengar[†], A Saurez, D Har . “**Few Contrastive Attention Heads Enable Visual Grounding in Large Vision-Language Models .**” *Transactions on Machine Learning Research (TMLR)*, 2026 .
- N Sengar[†], A Saurez, D Har . “**CAGSeg: Training-Free Referring Segmentation via Contrastive Attention Grounding in Vision-Language Models .**” *IEEE Signal Processing Letters*, 2026 .
- N Sengar, Y Lee, K Won Ko, D Har, S Lee[†] . “**Vision-Based Tire Lifting for Parking by Dual Parking Robot .**” *IEEE Sensors Journal*, 2025 .

Journal Articles

- N Sengar[†], I Kumari, J Lee, D Har . “**Blockchain-based Vision Transformer Network for Safe Driving Perception .**” *IEEE Sensors Journal*, 2024 . DOI
- S Mishra^{*†}, N Sengar^{*}, D Har . “**A Secure, Blockchain-Enabled Vehicular Sensor Communication Protocol With Deep Learning-Assisted Anomaly Detection .**” *IEEE Intelligent Transportation Systems Magazine*, 2024 . DOI
- Z Xu, N Sengar, T Chen[†], H Chung, O Oviedo-Trespalacios . “**Where is morality on wheels? Decoding large language model (LLM)-driven decision in the ethical dilemmas of autonomous vehicles .**” *Travel Behaviour and Society, Elsevier*, 2025 . DOI
- N Sengar, RC Joshi, MK Dutta[†], R Burget . “**EyeDeep-Net: A Multi-Class Diagnosis of Retinal Diseases using Deep Neural Network .**” *Neural Computing and Applications, Elsevier*, 2023 . DOI
- N Sengar, R Burget, MK Dutta[†] . “**A Vision Transformer Based Approach for Analysis of Plasmodium Vivax Life Cycle for Malaria Prediction Using Thin Blood Smear Microscopic Images .**” *Computer Methods and Programs in Biomedicine, Elsevier*, 2022 . DOI
- S Yadav, N Sengar, A Singh, A Singh, MK Dutta[†] . “**Identification of Disease using Deep Learning and Evaluation of Bacteriosis in Peach Leaf .**” *Ecological Informatics*, 2021 . DOI

Conference Papers

- J Lee, Q Lai-Dang, N Sengar, D Har† . “Robust Monocular Depth Estimation in Adverse Weather Conditions by Unsupervised Domain Adaptation .” *ECAI 2024* . [DOI](#)
- Z Xu, N Sengar, T Chen† . “Evaluation of Ethical Dilemmas in ChatGPT-Based Decision-Making of Autonomous Vehicles .” *The 91st Conference of Korean Society of Transportation* . [Link](#)
- N Sengar†, RC Joshi, MK Dutta . “An Efficient Artificial Intelligence-based approach for Diagnosis of Media Haze Disease .” *ICCCNT 2021, IEEE* . [DOI](#)
- N Sengar†, A Singh, V Yadav . “Classification of Documents using Bidirectional Long Short-Term Memory Recurrent Neural Network .” *ICSCSP20, Springer* . [Link](#)
- N Sengar†, A Singh, S Yadav, MK Dutta . “Automated System for Face-mask Detection using Convolutional Neural Network .” *ICMC, Springer* . [DOI](#)
- N Sengar†, N Fatima, V Yadav . “Auto Organizer: A Machine Learning based Tool for Automatic Organization of Files .” *International Conference on Signal Data Processing* . [DOI](#)
- N Sengar†, T Pan, S Das, KB Singh, SC Misra . “Pneumocystis Jirovecii Fungi Detection Using Microscopy Image: A Deep Learning Approach .” *ICAIE 2023, Springer* . [DOI](#)
- A Tiwari, N Sengar, V Yadav† . “Next Word Prediction Using Deep Learning .” *IEEE GUCON 2022* . [DOI](#)
- A Malviya, N Sengar, MK Dutta†, R Burget, V Myska . “Deep Learning Based Gastro Intestinal Disease Analysis Using Wireless Capsule Endoscopy Images .” *TSP 2022, IEEE* . [DOI](#)
- A Singh, N Sengar, MK Dutta†, R Burget . “Predicting Lung Cancer Through Histopathological Tissue Images .” *TSP 2022, IEEE* . [DOI](#)
- KB Singh†, N Sengar, D Das, SC Misra . “Village 5.0: Enabling Technologies and Its Applications in Development of Smart Village .” *IEEE iSES 2022* . [DOI](#)
- S Das†, KB Singh, N Sengar, T Pan, SC Misra . “AlphaAR: An Augmented Reality Based Intelligent Learning Platform for Preschool Children .” *ICAIE 2023, Springer* . [DOI](#)

Patent

- S. C. Misra, D. Das, V. Udutalapally, S. Misra, N Sengar, A. Ghosh . “Microcontroller Blockchain Network with Edge Cloud Interplay for Data Security from Source .” *Application No. 202211070221, Patent No. 529261,06-12-2022* .

RESEARCH EXPERIENCE

Graduate Research Assistant

Feb 2023 – Present

Vehicular Intelligence Lab, KAIST

- My dissertation shows that frozen LVLMs already encode spatial grounding knowledge in a small subset of attention heads, discoverable through contrastive prompting without any labeled data or fine-tuning.
- **Referring Expression Comprehension:** Training-free attention-head discovery framework; outperforms training-free baselines by up to 27.95% on RefCOCO.
- **Referring Expression Segmentation:** Extended to pixel-level masks via spectral label propagation—competitive with supervised methods, no external segmentation model.
- **Visual Question Answering:** Generalizing contrastive head discovery to open-ended visual reasoning.
- Developed a vision-based 6DOF pose estimation system for autonomous EV charging robot using object detection, PnP solving, and AprilTag-based localization.
- Developed a vision-based perception pipeline for a dual parking robot, including tire contact-point detection and camera-LiDAR steering angle estimation, achieving $\leq 2^\circ$ alignment error for autonomous alignment.
- Mentored 2 M.S. students.

Junior Research Fellow

June 2022 – Jan 2023

Indian Institute of Technology Kanpur (Host: Prof. Subhas Chandra Misra)

- Built a microcontroller blockchain network with edge-cloud interplay for secure agricultural data collection as part of the MEITY-funded "TribeConnect" smart tribal eco-platform project in Chhattisgarh.

Junior Research Fellow

Aug 2021 – May 2022

Centre for Advanced Studies (Host: Prof. Malay Kishore Dutta)

- Developed an AI-based assistive device for visually impaired users, generating real-time scene captions as part of a DST-funded research project.
- Developed a low-cost IoT-based intelligent stethoscope with embedded AI for early diagnosis of heart, lung, and prenatal conditions as part of a DST-funded project.

Artificial Intelligence Intern

May 2021 – Aug 2021

Farm365 Innovations (Startup)

- Developed a supervised learning-based neural network models for automated animal intrusion detection and deterrence in agricultural fields.

KIT Winter Internship Program

Dec 2019 – Jan 2020

Indian Institute of Technology Kanpur (Host: Dr. Vipul Arora)

- Implemented a query-by-humming song retrieval system using audio feature matching, in collaboration with a PhD researcher.

AWARDS & HONORS

2026	Nominated as an Alumni Member of the Board of Studies	Centre for Advanced Studies
2025	Selected for HKUST Overseas Study Program for Entrepreneurship Boot-camp	HKUST
2023	KAIST Graduate Scholarship (full funding, 4 years)	KAIST
2019	Awarded State Government Scholarship for M.Tech (2 years)	

TECHNICAL SKILLS

Programming	Python (advanced), Embedded C/C++, CUDA
ML / DL	Transformers, PyTorch, TensorFlow, Hugging Face Transformers
Computer Vision	OpenCV, 6DOF Pose Estimation, Open3D
Robotics Hardware	Raspberry Pi, Realsense D435i (depth camera), Sony IMX415, Livox Mid-70 LiDAR, ESP32 Microcontrollers, other utility sensors
Tools	Git, Docker, Linux, L ^A T _E X, Arduino IDE, Edge Impulse, Roboflow

PROFESSIONAL SERVICE

Reviewer (Top-Tier Venues)

- Journals: IEEE T-ITS (2026), IEEE Sensors (2023-2026), IEEE IoT (2024)
- Conferences: ICML (2026), IJCAI (2026), NeurIPS (2025), IV (2025)

Volunteer

- Student Ambassador in Here Comes KAIST (HCK) program, 2025-26
- Student Coordinator in KAIST International Summer School (KISS), 2024
- Student Mentor for the new Fall International Students, 2023, 2025
- Student Volunteer, International Conference on Contemporary Computing and Application (IC3A2020)
- Student Volunteer, Dr. Abdul Kalam Technical, Literary and Management Fest 2019-20 (State Level)

INVITED TALKS

2025	<i>Vision-Language Models for Scene Understanding in Autonomous Driving</i>	Workshop "AI in Robotics", Amity University
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